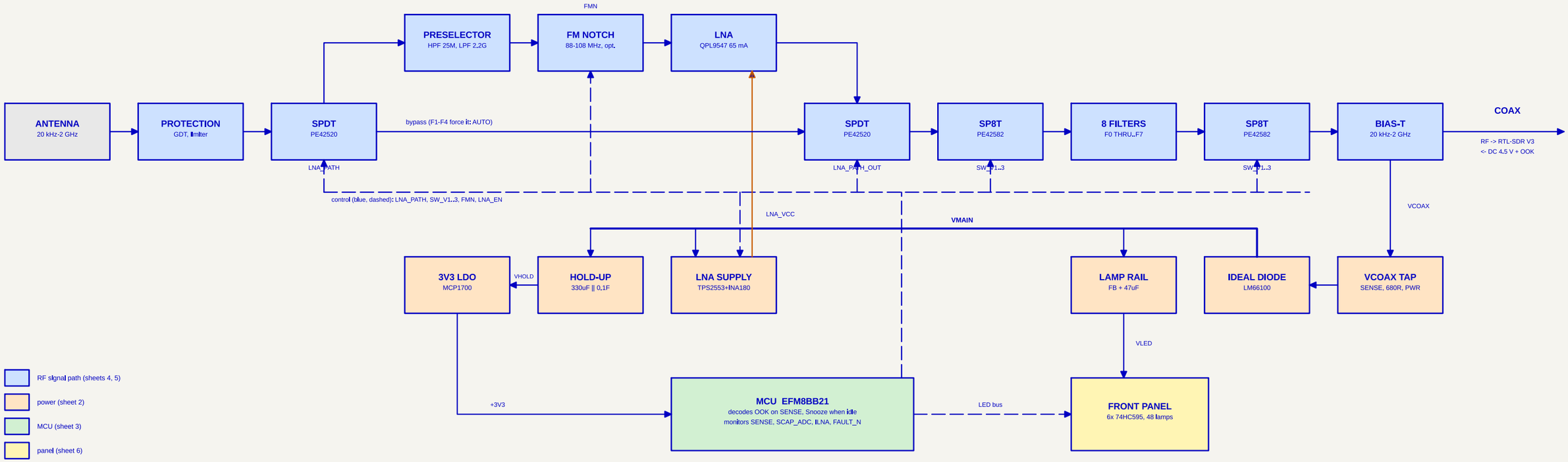
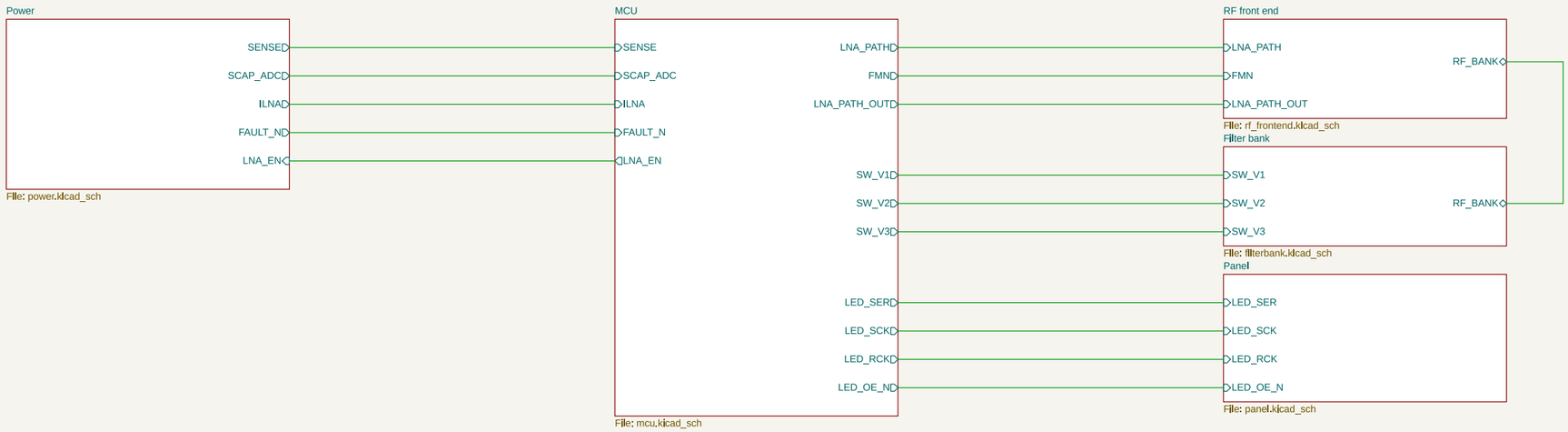
# Coaxolotl - block diagram




## Sheets



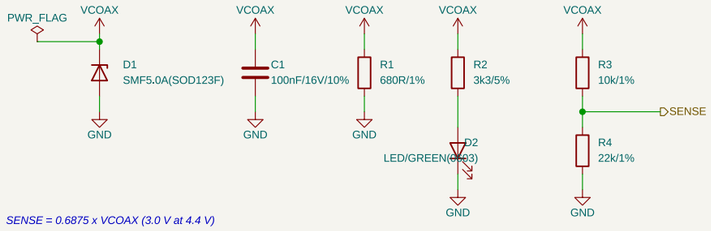

*Power and ground cross sheets only as power symbols (VCOAX, VMAIN, VHOLD, VLED, LNA_VCC, +3V3, GND).*

*Signals cross sheets as hierarchical labels through the sheet pins above. Firmware pin map: firmware/src/hal_efm8bb2.c*

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
Bias-T powered, OOK switched LNA + 8-way filter bank
**z2labs**

| | | |
|---|---|---|
| Sheet: / | | |
| File: coaxolotl.kicad_sch | | |
| ***Title: Coaxolotl*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/6 |

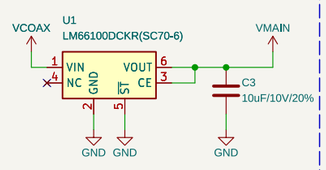
## VCOAX tap




*SENSE = 0.6875 x VCOAX (3.0 V at 4.4 V)*

*t_fall = 1.1 x R_eff x C_total: R1 sets the OOK falling edge*

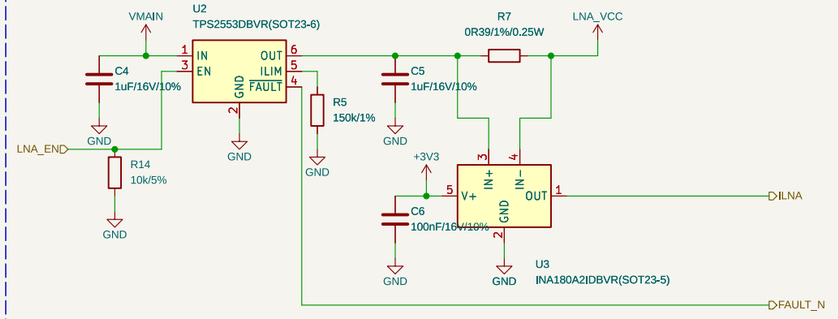
## Ideal diode




*CE = VOUT: comparator blocks*

*reverse current in OOK gaps*

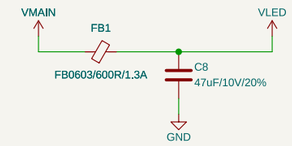
## LNA supply + current monitor




*Ilim = f(R5), TPS2553 Eq.1: VERIFY IOS(min) > 130 mA*

*ILNA = 0.39R x 50 V/V = 19.5 mV/mA (100 mA -> 1.95 V); Kelvin taps at R7 pads*

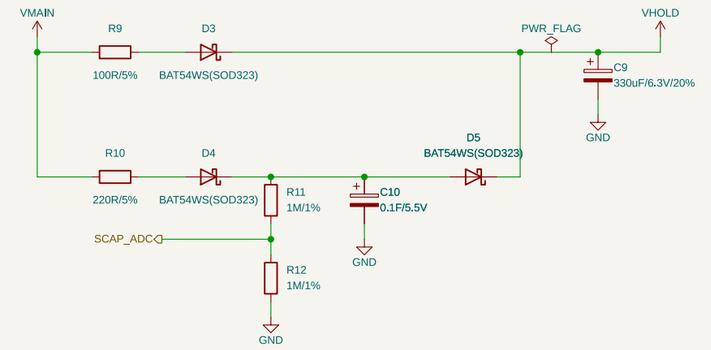
## Lamp rail




*48 lamps x 1 mA max on VMAIN:*

*dark in OOK gaps, no hold-up drain*

## Hold-up (diode-OR)

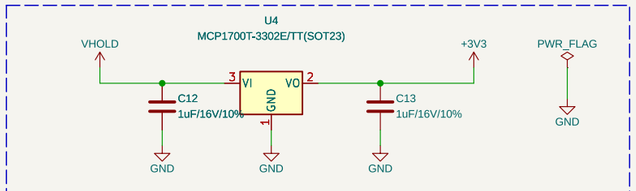



*fast path R9/D3: VHOLD up in ~30 ms (100R x 330uF) -> MCU boots*

*slow path R10/D4: tau ~22 s (220R x 0.1F); ride-through ~4 min at 0.6 mA*

*SCAP_ADC = VSCAP / 2 (HOLD lamp at >= 3.6 V)*

## 3V3 regulator




*MCP1700: Iq 1.6 uA, dropout 178 mV;*

*VO needs >= 1 uF ceramic*

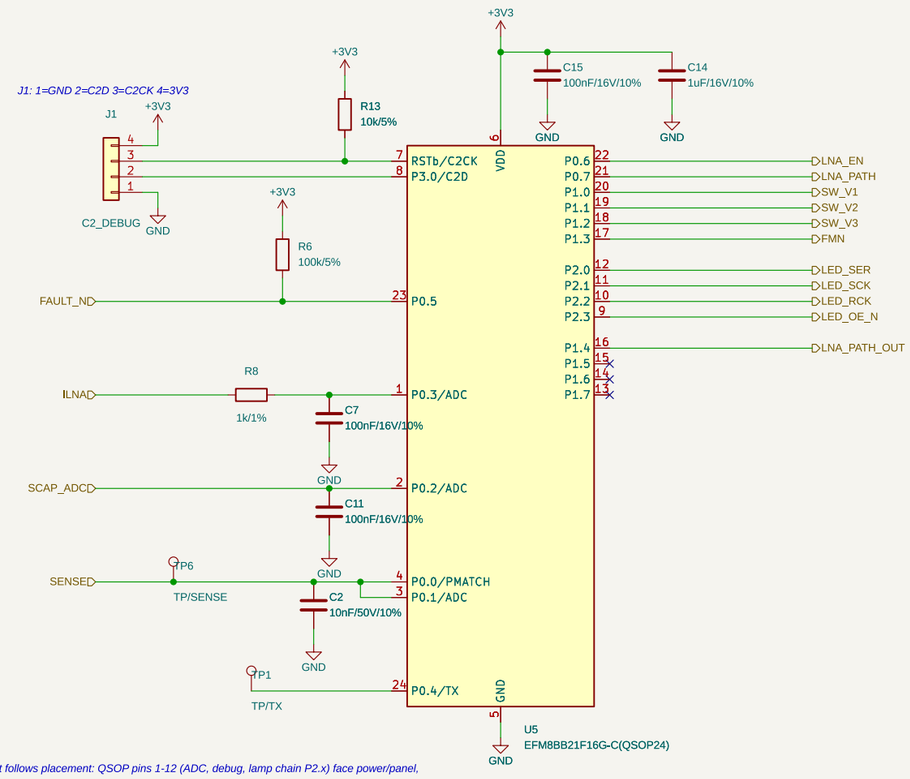
Generated by hardware/kicad/tools/build.py — edit the Python, not this file

VCOAX tap, ideal diode, LNA supply switch + current monitor, lamp rail, hold-up, 3V3

**z2labs**

| | | |
|---|---|---|
| Sheet: /Power/ | | |
| File: power.kicad_sch | | |
| ***Title: Power*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/6 |

## MCU




*Pinout follows placement: QSOP pins 1-12 (ADC, debug, lamp chain P2.x) face power/panel,*
*pins 13-24 (switch control, LNA_EN, FAULT_N) face the RF chain: no line crosses the MCU.*
*SYSCLK = HFOSC0/8 = 3.0625 MHz, Snooze when idle (no clock, no spurs). ILNA RC: fc = 1.6 kHz*

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
EFM8BB21F16G, C2 debug, ADC input conditioning
**z2labs**

| | | |
|---|---|---|
| Sheet: /MCU/ | | |
| File: mcu.kicad_sch | | |
| ***Title: MCU*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/6 |

## Antenna input + protection

J2
SMA/EDGE(ANT)

GDT1
4532-090V(1812)

D6
BAV99W(SOT323)

R23
100k/5%

*GDT 90 V, BAV99 low-C clamp, 100k bleed*

## LNA/BYP IN

C36
100nF/16V/10%

U6
PE42520C-Z(QFN16)

bypass: F1-F4 (HF/LF) always, LNA off or tripped

## LNA/BYP OUT

C37
100nF/16V/10%

U7
PE42520C-Z(QFN16)

RF_BANK

## Preselector (LNA path only)

*HPF: 5th-order Butterworth T, fc ~25 MHz*

*LPF: LTCC, fc ~2.2 GHz*

C24
200pF/50V/C0G

C25
68pF/50V/C0G

C26
200pF/50V/C0G

L1
220nH/5%

L2
220nH/5%

U12
LFCN-2250+(FV1206)

R22
0R/0603

## LNA

LNA_VCC

FB2
FB0603/600R/1.3A

L4
560nH/5%

C40
100pF/C0G

C32
100pF/C0G

R25
3k32/1%

C30
100pF/C0G

L8
18nH/2%

C28
1nF/50V/C0G

C31
1nF/50V/C0G

U13
QPL9547TR7(DFN8-2x2)

*SWL434 stage 1*
*+ 18 nH (EVB L1)*
*1 nF DC blocks*

## FM notch (DNP option)

U9
NA(PE42520C-Z)

L3
NA(56nH/2%)

C27
NA(47pF/C0G)

U10
NA(PE42520C-Z)

*tank 56 nH || 47 pF: f0 = 98 MHz*

## Control inputs (reset-safe pull-downs)

FMN
R19
10k/5%

LNA_PATH
R15
10k/5%

LNA_PATH_OUT
R26
10k/5%

*LNA_PATH (U6) and LNA_PATH_OUT (U7, own GPIO): 0 = bypass, 1 = LNA. FMN 0 = through. U7/U10: LS = GND.*

*PE42520 RF ports sit at 0 VDC: no blocking caps except around the LNA.*

Generated by hardware/kicad/tools/build.py — edit the Python, not this file
Antenna protection, LNA/bypass select, preselector, FM notch option, LNA

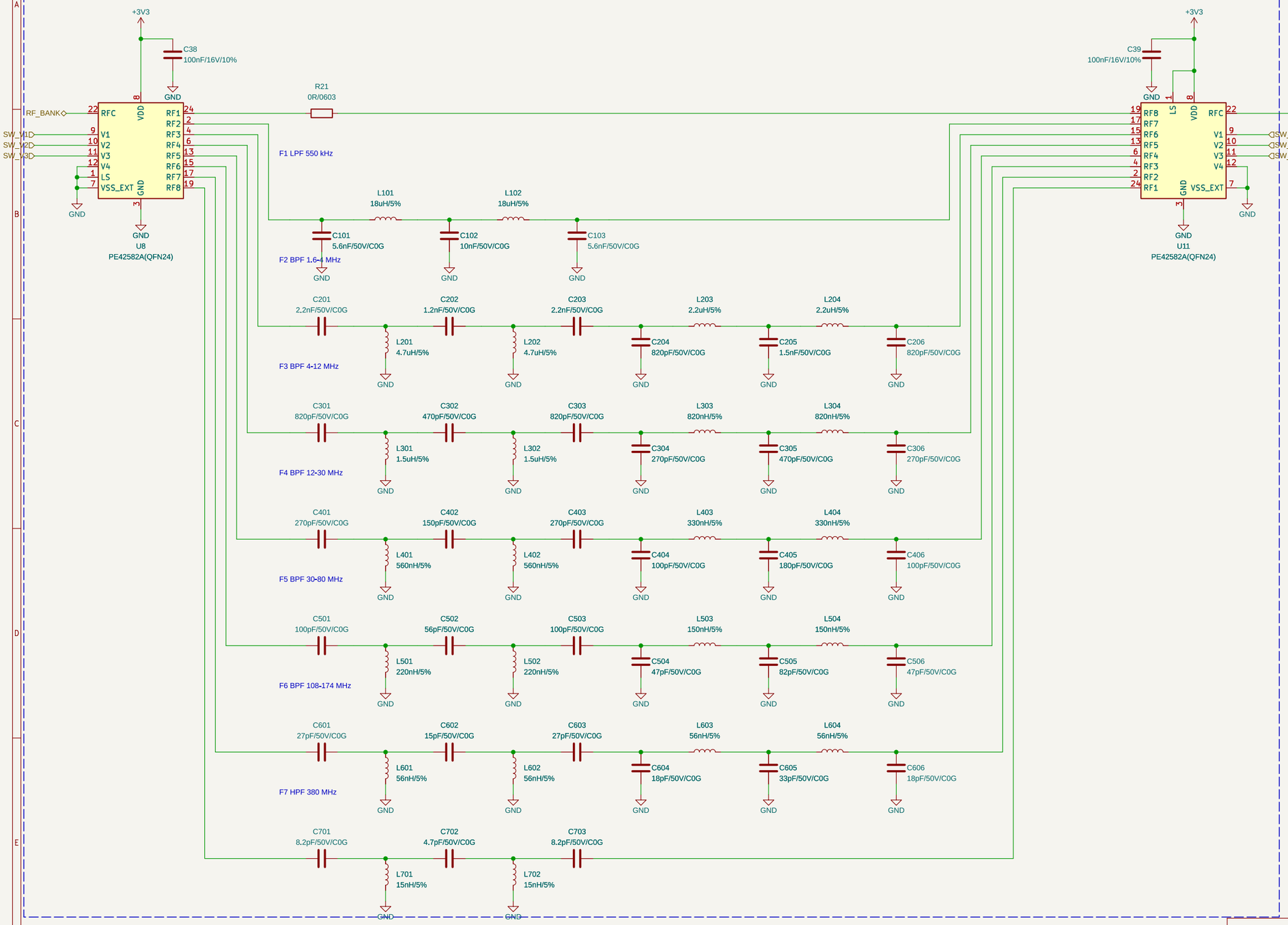
**z2labs**

Sheet: /RF front end/
File: rf_frontend.kicad_sch

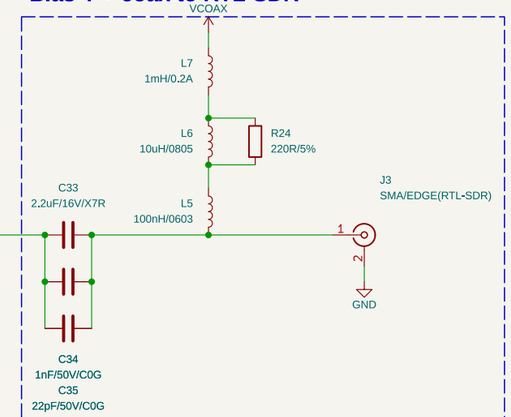
**Title: RF front end**

| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/6 |

## Filter bank

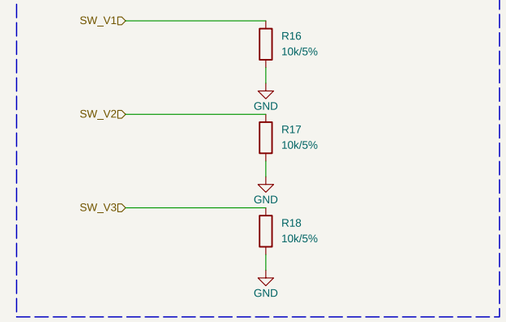



## Bias-T + coax to RTL-SDR




*DC block 2.2 uF ‖ 1 nF ‖ 22 pF; DC feed 100 nH -> 10 uH ‖ 220R -> 1 mH:*
*20 kHz-2 GHz. VERIFY S21/S11 with a VNA. D1 (TVS) sits at J3 (power sheet).*

## Control inputs

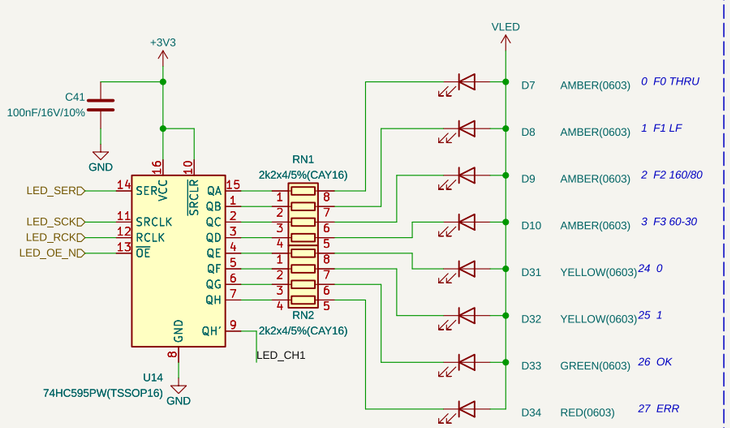



*Branch k: U8 RF(k+1) (LS = 0), U11 RF(8-k) (LS = 1, mirror port).*
*Same V lines on both. U8 code (V3 V2 V1):*
*RF1 000, RF2 100, RF3 010, RF4 110, RF5 001, RF6 101, RF7 011, RF8 111*
*(mapped in tools/gen_bandplan.py). V4 = LS = VSS_EXT = GND.*
*Filters: 5th-order Chebyshev 0.1 dB, 50 ohm, E12 values,*
*BPF = HPF + LPF (tools/filters.py). Ideal-part response there;*
*VERIFY with real L Q / SRF and the layout on a VNA.*

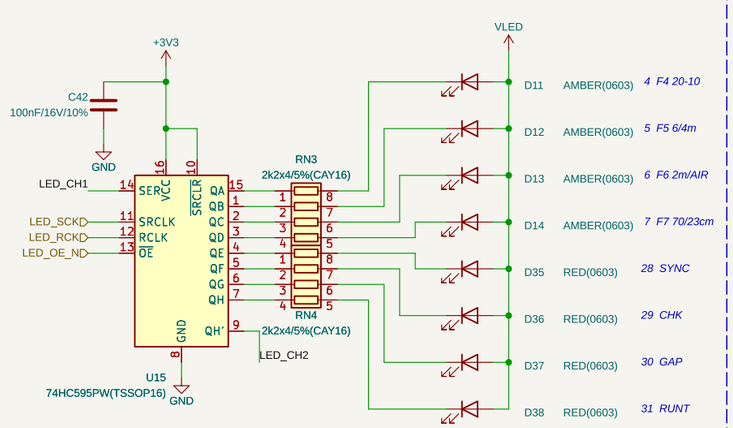
Generated by hardware/kicad/tools/build.py — edit the Python, not this file
2x PE42582 SP8T, 8 filter branches, bias-T, coax
**z2labs**

| | | |
|---|---|---|
| Sheet: /Filter bank/ | | |
| File: filterbank.kicad_sch | | |
| ***Title: Filter bank*** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/6 |

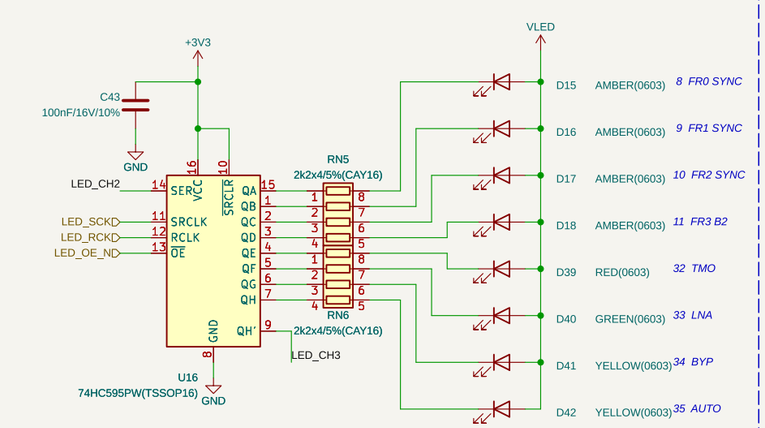
## columns 0-3 (lamps 0-3, 24-27)




## columns 4-7 (lamps 4-7, 28-31)




## columns 8-11 (lamps 8-11, 32-35)




## columns 12-15 (lamps 12-15, 36-39)



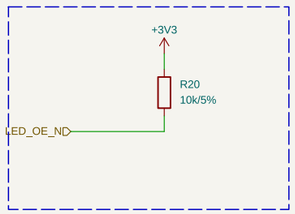

## columns 16-19 (lamps 16-19, 40-43)



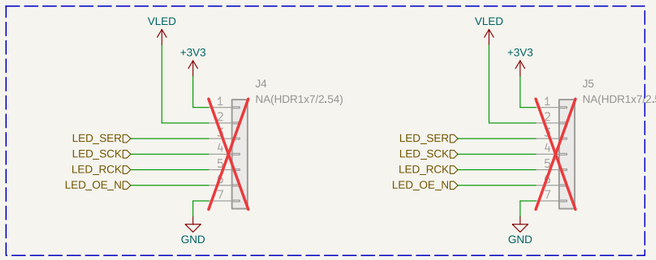

## columns 20-23 (lamps 20-23, 44-47)




## OE_N pull-up




## Snap-off lamp strip link




*Lamp n: row n/24, column n%24; register (n%24)/4 (U14 first), output (n%4) + 4*(n/24) - hal_panel.*

*Q low = lamp on (sink). 1 mA per lamp: (VLED - 2 V - 0.1 V) / 2k2. Use 2 V LEDs only.*

*Lamp test lights all 48 (48 mA) with the LNA still off. Outside it, at most 23 lamps.*

*Panel modes (config frame, kept in flash): full / bands = F0-F7 only / off = OE_N high.*

*U14-U19, RN1-RN12, D7-D54, C41-C46 and J5 sit on the snap-off strip; J4 on the main board.*

Generated by hardware/kicad/tools/build.py — edit the Python, not this file

48-lamp front panel: 6x 74HC595 sinking, 2k2 arrays, VLED

**z2labs**

| | | |
|---|---|---|
| Sheet: /Panel/ | | |
| File: panel.kicad_sch | | |
| **Title: Panel** | | |
| Size: A3 | Date: 2026-10-26 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 6/6 |